

HUBERT KIM

AUTOMATION RESEARCH ENGINEER

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FAVORITE TOOLS

Camera

- 2D Optical Camera
- RGB-D 3D Camera
- Laser Profilometer

Simulation & Design

- SolidWorks / MATLAB

Human-Machine Interface

- TCP/IP via Python

SOFT SKILL SET

Action Planning Skill

- Deriving long/short-term approaches
- Most Viable Product Test

Debugging Skill

- Error boundary control
- Operational safety margin
- Risk Assessment Tool

EDUCATION

PhD,

Mechanical Engineering
Virginia Tech, Blacksburg, VA
Earned in Dec 2021

BS, *cum laude*,

Mechanical Engineering
NYU Tandon, Brooklyn, NY
Earned in May 2015

SUMMARY

Data-driven robotics engineer specializing in automating complex human motion.

- 3 years of experience in an early failure & early learn project (TRL < 5)
- Setting up robot perception via environment registration
- Defining application-specific boundary conditions for gripping strategy & motion planning in creating/breaking physical contact
- Refining precision to reduce simulation-to-real gap

PROFESSIONAL EXPERIENCE

AUTOMATION/MECHATRONICS ENGINEER

Aug 2022 – Current

Mechanical Components and Systems Lab | GE Aerospace Research Center

Product cost saving and part assembly quality improvement by automating the manufacturing system.

- Established a multi-robot cell with an HMI platform, providing coordinated robot motion with an offline trajectory planning feature.
- Identified and analyzed risks during pick and place tasks to reduce the probability and impact of the root causes.
- Prototyped a surveillance robot to solve the accessibility problem under a confined space while keeping its structural rigidity (IP filed & Patent Award)
- Earned a Spotlight award for reaching a significant 1st milestone on the project by delivering and demonstrating the high-quality automation result on time – Apr 2025
- Earned an Applause award for supporting the offsite leadership tour – Aug 2024
- Earned a Bravo award for demonstrating real grit in the innovation of robotic manufacturing. This demonstration showed the feasibility to the Aerospace senior leadership and was recognized as an essential technology for the program – Oct 2023

GRADUATE RESEARCH ASSISTANT

May 2015 – Dec 2021

Assistive Robotics Laboratory | Virginia Tech

Advisor: Dr. Alan T. Asbeck

The physiological states of the wearer from the exoskeleton and the robot's perception were studied to understand the user's intention.

Dissertation: "Joint Torque Feedback for Arm Motion Training"

UNDERGRADUATE RESEARCH ASSISTANT

May 2013 – Dec 2015

Dynamic System Laboratory | NYU

Advisor: Dr. Maurizio Porfiri

System identification: understand and apply frequency domain modeling theory to real-world applications.

Thesis: "Voltage attenuation along the electrodes of ionic polymer metal composites"